H03e - A rotating Smartphone as an accelerated Frame of Reference

Group 6: Laiba Rehan Zahid - 3771917 (50%), Syed Muhammad Haider Amir - 3787940 (50%)

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1 Introduction

In this experiment, the motion of a rotating smartphone is investigated to study the behavior of physical quantities in an accelerated frame of reference. When a body rotates, the reference frame attached to it is non-inertial, and additional apparent (fictitious) forces such as the centrifugal and Euler forces must be considered to describe the observed accelerations. Using the sensors integrated in a modern smartphone, specifically the gyroscope and the accelerometer, these effects can be measured and analyzed in detail.

The main objective of this experiment is to record the angular velocity and acceleration of a smartphone set into rotation on a horizontal surface, and to use these measurements to determine the position of the acceleration sensor inside the device. The experiment combines theoretical understanding of accelerated frames with practical data analysis methods such as curve fitting and symbolic regression to identify the sensor location with quantitative accuracy.

2 Theoretical Analysis

2.1 Accelerated Frames of Reference

In classical mechanics, Newton's laws of motion are valid only in inertial reference frames, those that are not accelerating with respect to distant fixed stars. If an observer is located in a reference frame that itself accelerates or rotates, the frame is termed a *non-inertial* or *accelerated frame of reference*. In such frames, the apparent motion of objects cannot be explained solely by real external forces. Additional apparent forces, known as *fictitious forces*, must be introduced to preserve the form of Newton's second law.

When the reference frame rotates with an angular velocity vector $\vec{\omega}$, the acceleration $\vec{a}'$ of a point with position vector $\vec{r}$ relative to the rotating frame is related to its acceleration $\vec{a}$ in the inertial laboratory frame by

$$\vec{a}' = \vec{a} - \vec{g} = -\vec{\omega} \times (\vec{\omega} \times \vec{r}) - 2\,\vec{\omega} \times \vec{v}' - \vec{\alpha} \times \vec{r}, \tag{1}$$

where $\vec{v}'$ is the velocity in the rotating frame and $\vec{\alpha} = \dot{\vec{\omega}}$ is the angular acceleration of the frame.

2.2 Fictitious Forces in a Rotating Frame

The three fictitious accelerations that appear in Eq. (1) correspond to distinct apparent forces:

- Centrifugal acceleration:
$$\vec{a}_{\mathrm{cf}} = -\,\vec{\omega} \times (\vec{\omega} \times \vec{r}), \tag{2}$$
  which acts radially outward from the rotation axis and increases with the distance r from the axis and the square of the angular velocity magnitude.
- Coriolis acceleration:
$$\vec{a}_{\mathrm{c}} = -\,2\,\vec{\omega} \times \vec{v}', \tag{3}$$
  which arises when a mass moves with velocity $\vec{v}'$ relative to the rotating frame. It acts perpendicular to both $\vec{\omega}$ and $\vec{v}'$ and is responsible for deflection effects such as those observed in large-scale atmospheric motion.
- Euler acceleration:
$$\vec{a}_{\alpha} = -\,\vec{\alpha} \times \vec{r}, \tag{4}$$
  which occurs when the angular velocity of the frame changes in time, i.e. during non-uniform rotation.

2.3 Simplifications for the Present Experiment

In this experiment, the smartphone rests on a horizontal surface and rotates about a nearly vertical axis without any translational motion of the sensor relative to the phone. Therefore, $\vec{v}' = 0$, and the Coriolis acceleration vanishes:

$$\vec{a}_{\mathrm{c}} = 0. \tag{5}$$

Only the centrifugal and Euler terms contribute to the total apparent acceleration.
The total acceleration measured by the smartphone's accelerometer is thus given by

$$\vec{a} = \vec{g} + \vec{a}_{\mathrm{cf}} + \vec{a}_{\alpha} = \vec{g} - \vec{\omega} \times (\vec{\omega} \times \vec{r}) - \vec{\alpha} \times \vec{r}. \tag{6}$$

Where:

$\vec{a}$: Acceleration
$\vec{g}$: Gravitational Acceleration
$\vec{\omega}$: Angular Velocity
$\vec{\alpha}$: Angular Acceleration
$\vec{r}$: Vector from the Contact Point to the Position of the Sensor

2.4 Measured Quantities and Data Evaluation Goal

The Phyphox application allows simultaneous recording of the following quantities:

- Angular velocity $\vec{\omega}$ from the gyroscope,
- Acceleration $\vec{a}$ from the accelerometer, and
- Linear acceleration $\vec{a}_{\mathrm{lin}}$, which is the acceleration without the gravitational contribution, computed as

$$\vec{a}_{\mathrm{lin}} = \vec{a} - \vec{g}. \tag{7}$$

By comparing the measured $\vec{a}_{\mathrm{lin}}(t)$ with the theoretical expression derived from Eq. (6), the position vector $\vec{r}$ of the acceleration sensor within the smartphone can be obtained using curve fitting and symbolic regression methods. This provides a direct experimental connection between the dynamics of rotating frames and the internal geometry of the device.

3 Experimental Analysis

3.1 Experimental Setup

The experiment was carried out using a Google Pixel 7 smartphone as the rotating body and measurement device. The device has a mass of $m = 197$ g and dimensions of 15.56 cm × 7.32 cm × 0.87 cm. The built-in sensors of the smartphone were accessed using the application Phyphox, which allows direct readout of gyroscope and accelerometer data, as well as real-time data transfer to a laptop via remote access.
The following quantities were recorded simultaneously:

- Angular velocity $\vec{\omega}$ (rad/s) from the gyroscope,
- Acceleration $\vec{a}$ (m/s^2) from the accelerometer, and
- Linear acceleration $\vec{a}_{\mathrm{lin}}$ (m/s^2), i.e. acceleration without gravity.

Five separate measurement runs were performed, each consisting of complete rotations of the smartphone around a vertical axis. The smartphone was placed flat on a smooth horizontal surface and rotated manually in a counterclockwise direction. To prevent scratches or damage to either the smartphone or the surface, a thin sheet of paper was placed in between. Before starting each measurement, the smartphone was kept stationary to record a short baseline period, ensuring the sensors were calibrated and stationary offsets could be identified. All rotations were conducted at approximately the same angular speed to maintain consistency between runs.

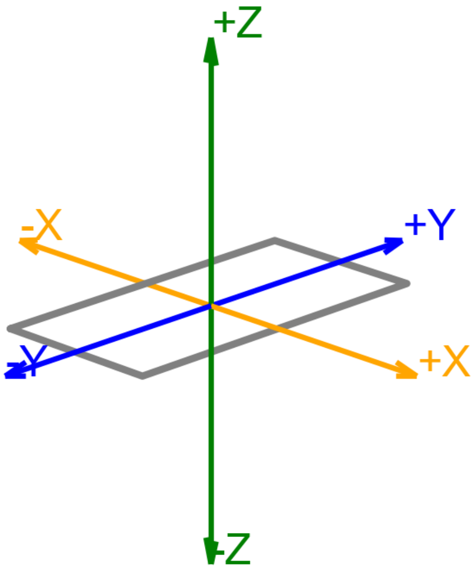

Figure 1: Coordinate System of Smartphone

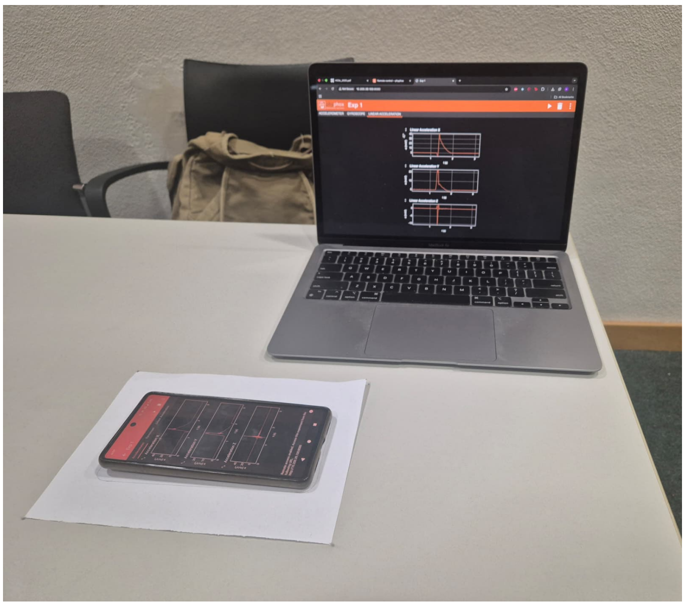

Figure 2: Experimental Setup

3.2 Data Acquisition

Data collection was conducted using the Phyphox remote access feature, which allowed the smartphone to transmit real-time sensor data to a connected laptop over a local network. This setup minimized handling errors and unintentional vibrations, as the smartphone did not need to be touched during recording. The laptop interface was used to start and stop measurements precisely and to save data directly in `.csv` format for further processing.

The recorded quantities included the three Cartesian components of angular velocity $(\omega_x, \omega_y, \omega_z)$, acceleration (a_x, a_y, a_z), and linear acceleration $(a_{\mathrm{lin},x}, a_{\mathrm{lin},y}, a_{\mathrm{lin},z})$. To reduce measurement noise and sensor drift, each dataset was trimmed to exclude the stationary periods before and after rotation. A total of five datasets were recorded and compared to verify reproducibility. The best-quality run, exhibiting smooth, constant angular velocity, was selected for detailed analysis and fitting.

3.3 Data Visualization

The recorded datasets for angular velocity, acceleration, and linear acceleration were visualized using `Python` and `Matplotlib`. Each component was plotted as a function of time to evaluate data quality and to identify the dynamic behavior during the smartphone's rotation. All figures include gridlines, labeled axes, and legends for clarity.

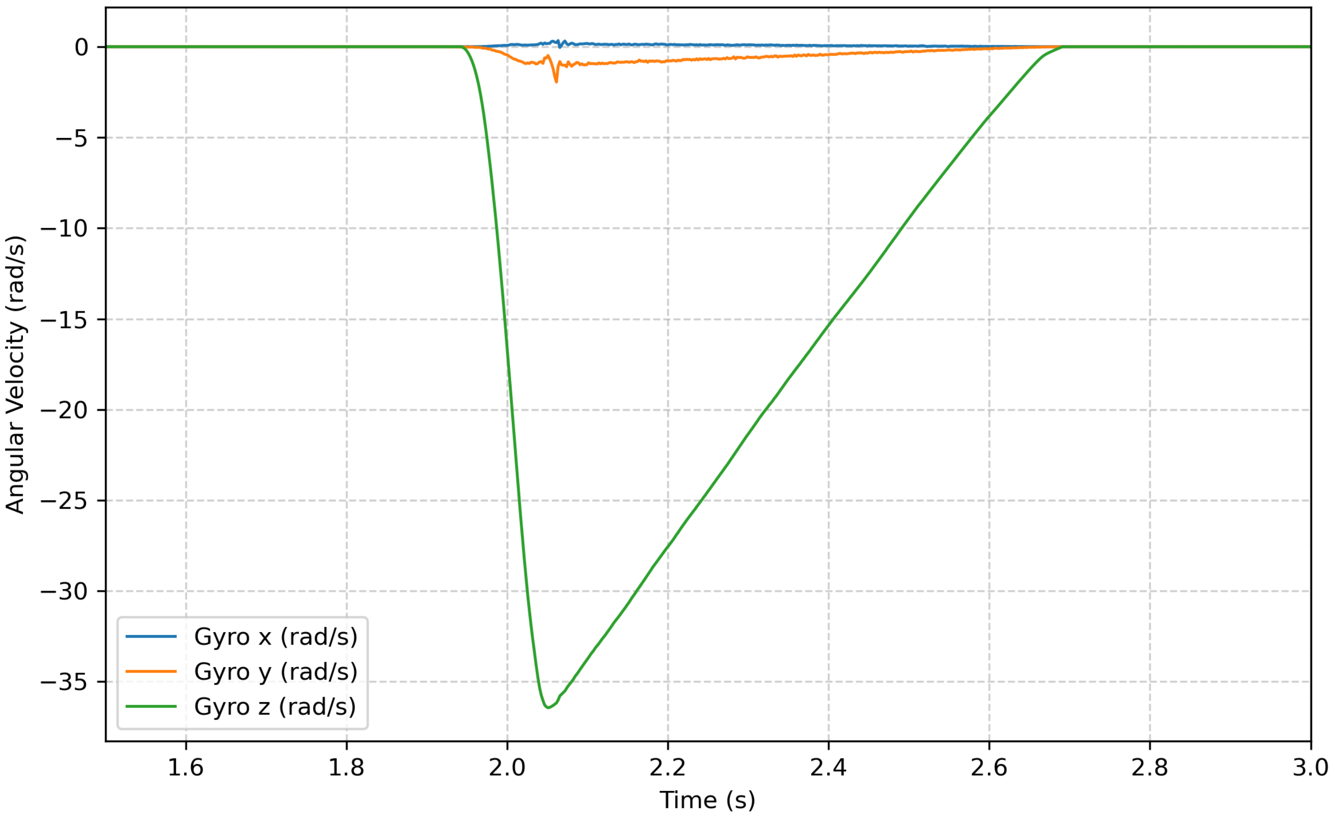

Figure 3: Velocity Components vs Time

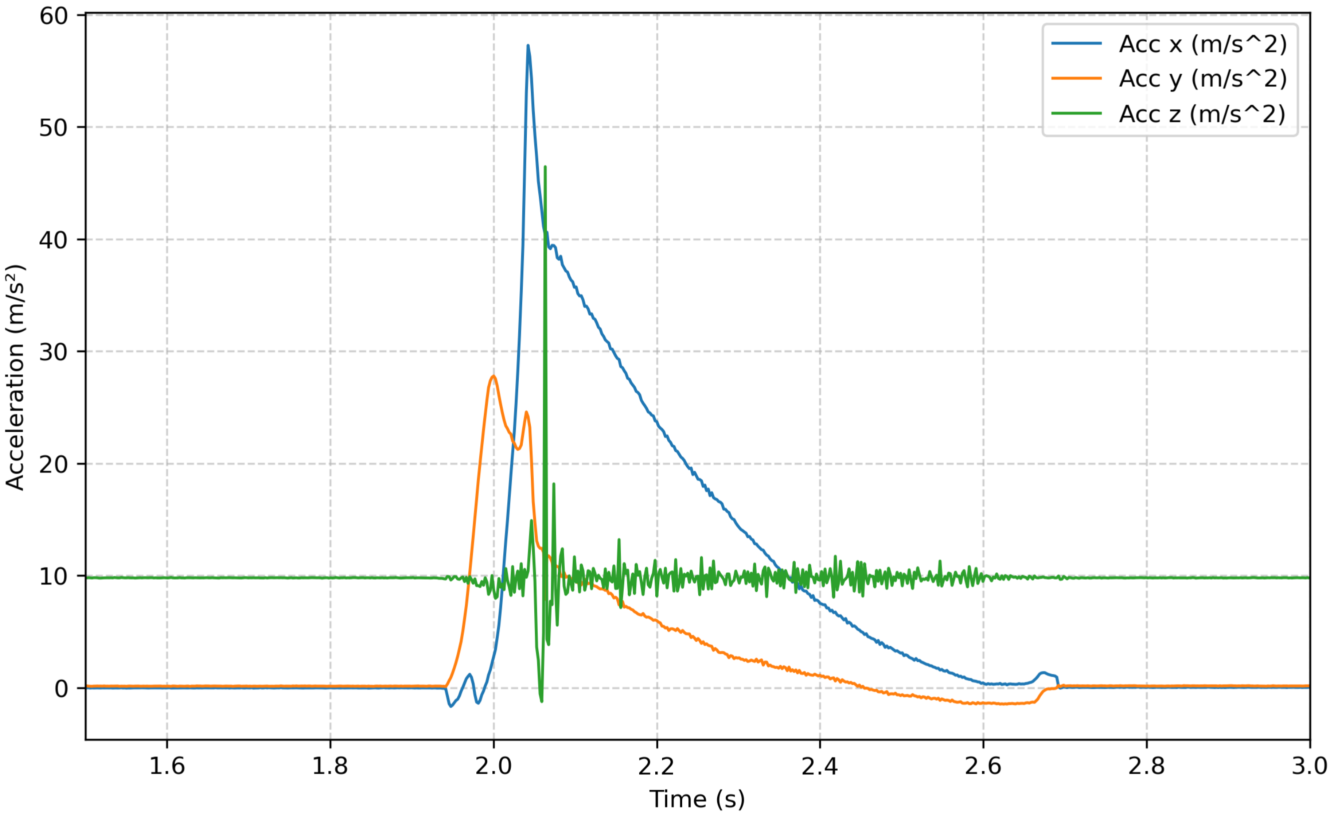

Figure 4: Acceleration Components vs Time

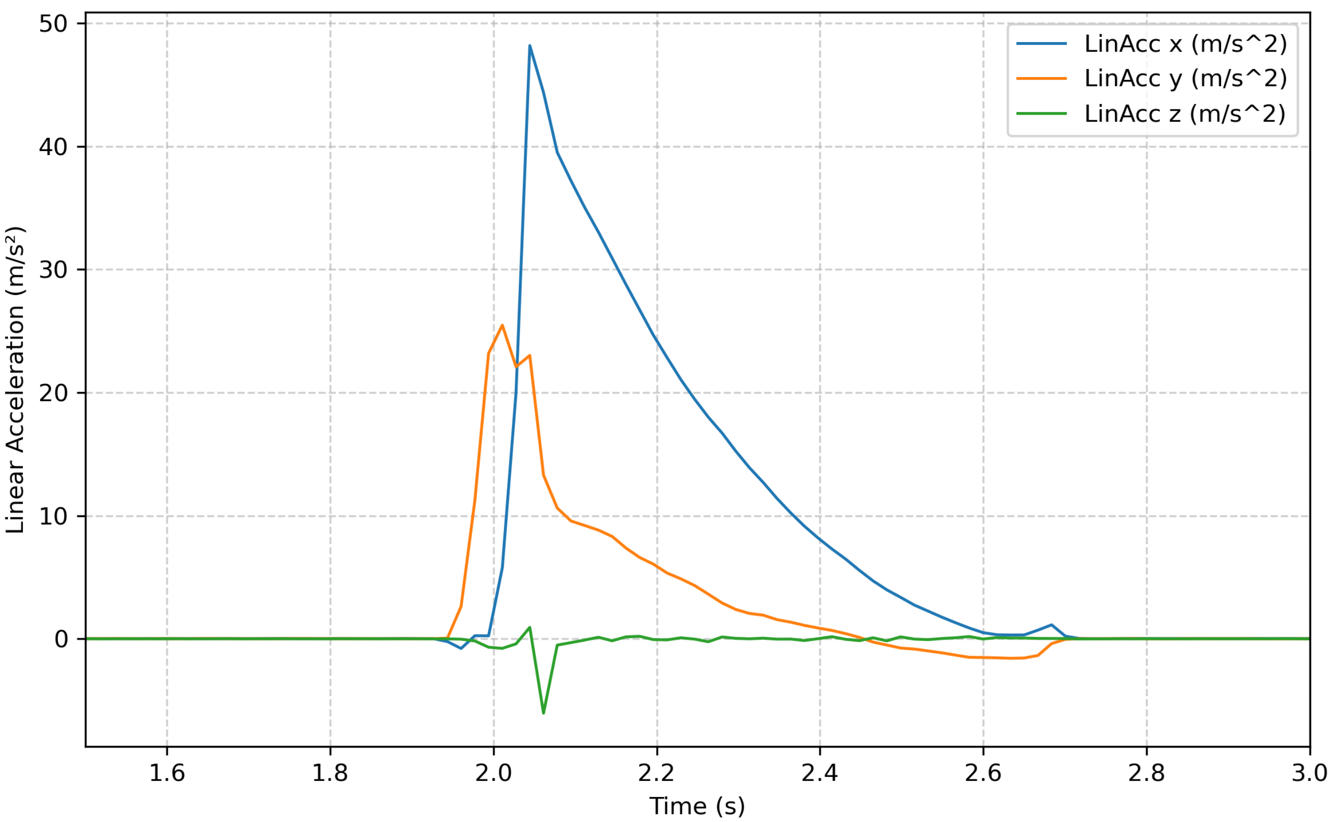

Figure 5: Linear Acceleration Components vs Time

Overall, the plots show consistent and physically reasonable trends. The dominant rotation about the z-axis is evident from the gyroscope data, while the accelerometer and linear acceleration components exhibit the expected sinusoidal variations due to the centrifugal and Euler accelerations in the rotating frame. These visualizations confirm the quality of the recorded data and provide the basis for the subsequent fitting and regression analysis of the sensor position.

4 Curve Fitting

4.1 Theoretical Background

The position of the smartphone's acceleration sensor can be determined from the relation between the measured linear acceleration and the rotational motion of the device. According to the theory of rotating frames (see Eq. (6)), the acceleration in the smartphone's frame of reference is given by

$$\vec{a}_{\text{lin}} = -\vec{\omega} \times (\vec{\omega} \times \vec{r}) - \vec{\alpha} \times \vec{r}, \tag{8}$$

where $\vec{\omega}$ is the angular velocity, $\vec{\alpha}$ the angular acceleration, and $\vec{r}$ the vector from the rotation axis to the position of the acceleration sensor.

Since the smartphone rotates about its z-axis while lying flat on a horizontal surface, the angular velocity and acceleration can be written as

$$\vec{\omega} = (0,\, 0,\, \omega_z), \qquad \vec{\alpha} = (0,\, 0,\, \dot{\omega}_z).$$

Inserting these into Eq. (8) yields the simplified planar model:

$$\begin{aligned} a_{\text{lin},x} &= \omega_z^2\, r_x + \dot{\omega}_z\, r_y, \\ a_{\text{lin},y} &= \omega_z^2\, r_y - \dot{\omega}_z\, r_x, \\ a_{\text{lin},z} &\approx 0. \end{aligned} \tag{9}$$

This system relates the measured linear accelerations to the angular velocity $\omega_z(t)$, its time derivative $\dot{\omega}_z(t)$, and the unknown coordinates r_x and r_y of the sensor.

4.2 Method and Implementation

The fitting procedure was implemented in `Python` using the `SciPy` library. The gyroscope data $\omega_z(t)$ were first numerically differentiated to obtain the angular acceleration $\dot{\omega}_z(t)$. Both quantities were interpolated to match the time points of the linear acceleration data.

The two-dimensional model of Eq. (9) was fitted to the data using a non-linear least-squares minimization, minimizing the sum of squared residuals:

$$\chi^2 = \sum_t \left[(a_{\text{lin},x}^{\text{meas}} - a_{\text{lin},x}^{\text{model}})^2 + (a_{\text{lin},y}^{\text{meas}} - a_{\text{lin},y}^{\text{model}})^2 \right]. \tag{10}$$

The covariance matrix of the fit parameters was derived from the Jacobian of the residual function, providing the statistical uncertainties of r_x and r_y.

4.3 Results

The least-squares fitting yielded the following coordinates for the acceleration sensor relative to the rotation axis:

$$\begin{aligned} r_x &= (32.01 \pm 0.17)\ \text{mm}, \\ r_y &= (11.09 \pm 0.17)\ \text{mm}. \end{aligned}$$

These values indicate that the acceleration sensor is located approximately 3.2 cm along the x-direction and 1.1 cm along the y-direction from the rotation axis, consistent with the expected placement of sensors near the central region of the device. The small uncertainties (on the order of 0.17 mm) confirm the robustness of the fit and the low level of noise in the measured data.

4.4 Uncertainty Analysis

Uncertainties in the fitted coordinates arise from several sources:

- **Sensor noise:** Random fluctuations in the accelerometer and gyroscope readings introduce small variations in ω_z and a_{lin}.
- **Numerical differentiation:** The computation of $\dot{\omega}_z$ amplifies high-frequency noise in $\omega_z(t)$.
- **Mechanical imperfections:** Slight tilts, uneven surfaces, or off-center rotation may lead to systematic errors.

The statistical uncertainties of r_x and r_y were obtained from the covariance matrix of the least-squares fit,

$$\sigma_{r_i} = \sqrt{C_{ii}},$$

where C_{ii} are the diagonal elements of the covariance matrix. The resulting relative uncertainties are below 1%, which is well within the precision limits expected for high-quality smartphone sensors. Systematic uncertainties, primarily due to imperfect alignment and frictional effects, are estimated to contribute an additional error of a few tenths of a millimeter but do not significantly affect the overall conclusions.

4.5 Measured and Fitted Data

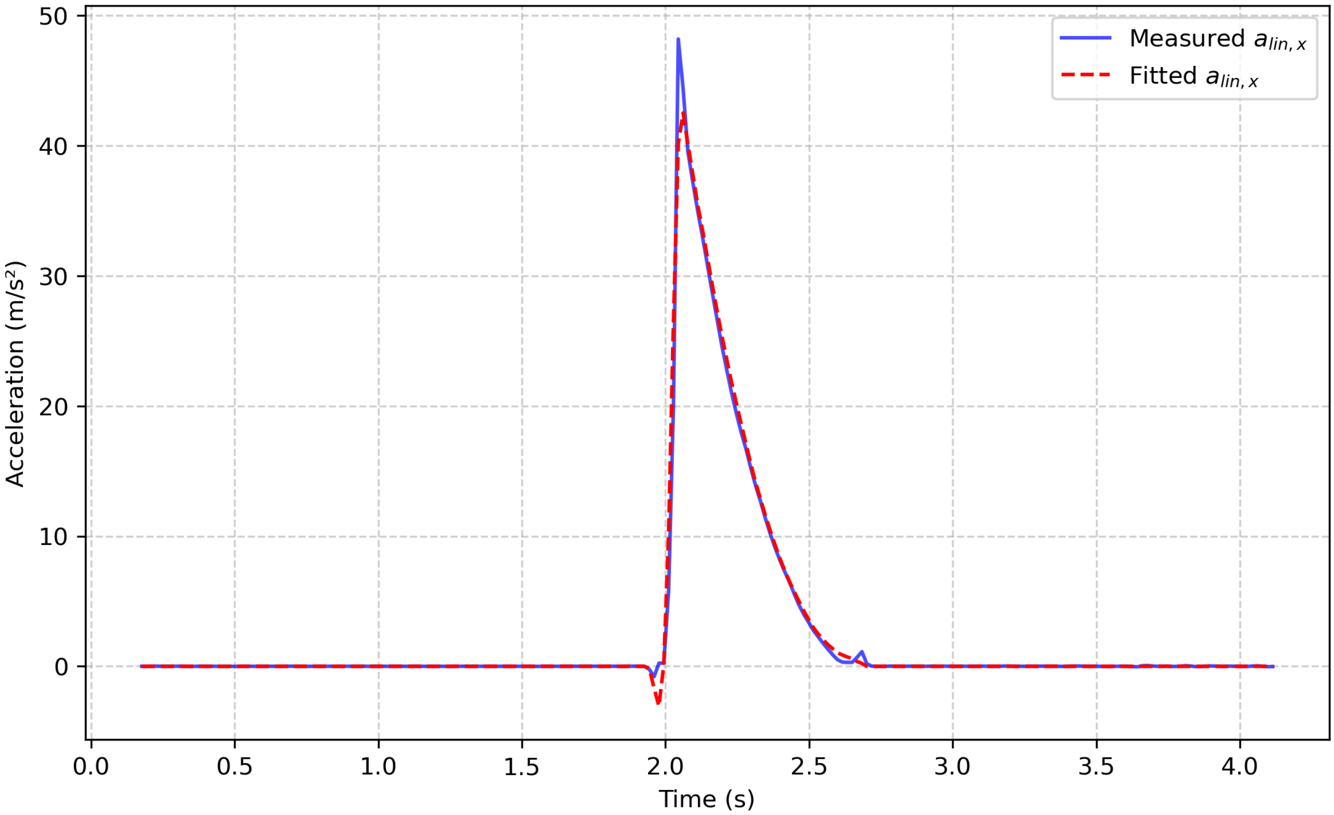

Figure 6: Measured vs Fitted Linear Acceleration (x-component)

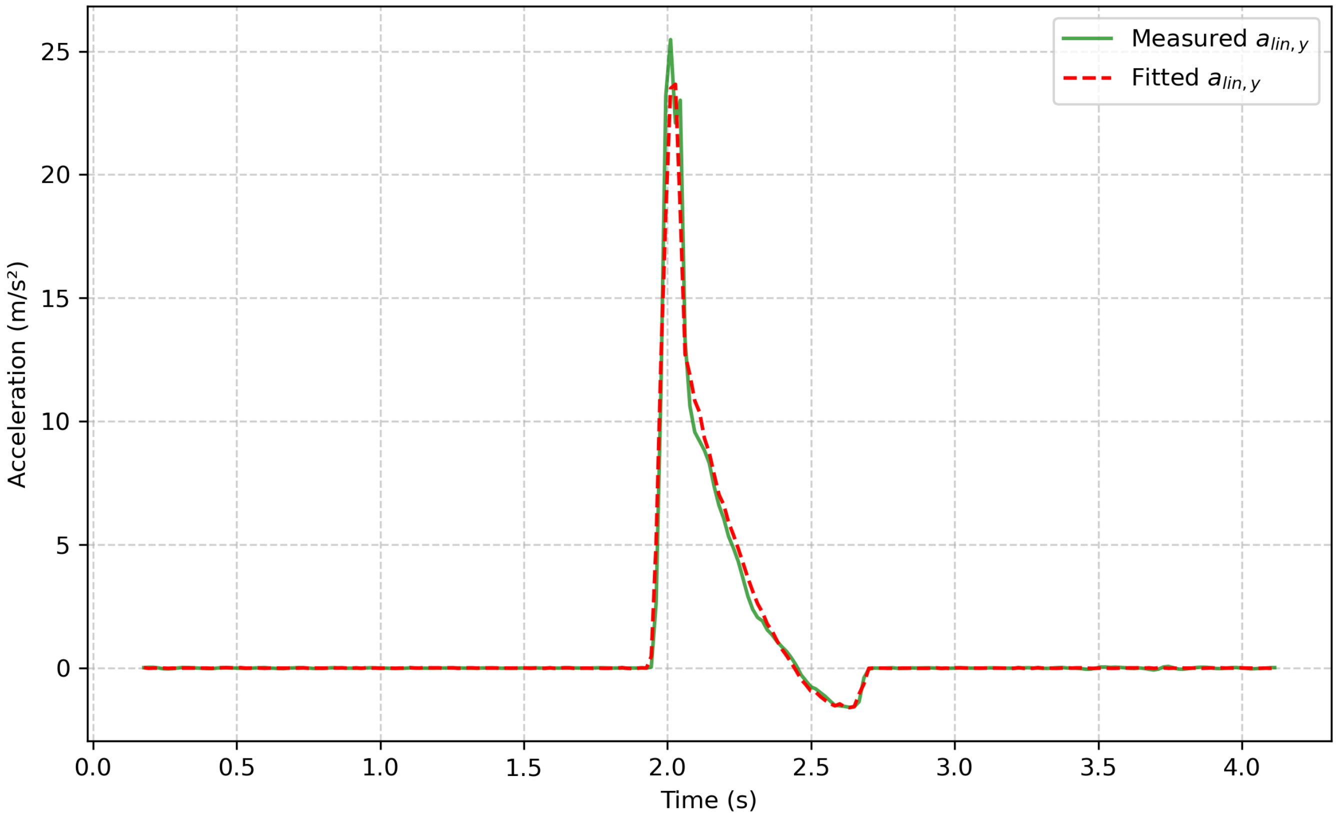

Figure 7: Measured vs Fitted Linear Acceleration (y-component)

Both fitted components show excellent agreement with the experimental data. Residuals remain within the expected noise range of the smartphone sensors, validating the reliability of the model and confirming the determined sensor position.

5 Symbolic Regression

5.1 Theoretical Background

In the previous section, the position of the smartphone's acceleration sensor was determined using a model-based curve fitting approach. To verify these results independently, a data-driven method based on *symbolic regression* was applied. Symbolic regression aims to automatically discover analytical expressions that best describe a dataset, without prior assumptions about the specific mathematical form. In this experiment, the method searches for equations that relate the measured angular velocity $\omega_z(t)$, its time derivative $\dot{\omega}_z(t)$, and the linear acceleration components $a_{\mathrm{lin},x}$ and $a_{\mathrm{lin},y}$. According to the theoretical model of motion in a rotating frame, the linear accelerations are related to the sensor coordinates (r_x, r_y) by:

$$\begin{aligned} a_{\mathrm{lin},x} &= \omega_z^2\, r_x + \dot{\omega}_z\, r_y, \\ a_{\mathrm{lin},y} &= \omega_z^2\, r_y - \dot{\omega}_z\, r_x. \end{aligned} \tag{11}$$

Symbolic regression was used to rediscover these relations directly from the measured data and to estimate the coefficients corresponding to r_x and r_y.

5.2 Method and Implementation

The symbolic regression was performed in `Python` using the `PySR` library, which evolves candidate symbolic expressions through a genetic programming algorithm. The gyroscope data provided $\omega_z(t)$, while $\dot{\omega}_z(t)$ was obtained by numerical differentiation and interpolated to the time grid of the linear acceleration data.

The symbolic regression model was configured to search for analytical relationships of the form

$$a_{\mathrm{lin},i} = f(\omega_z, \dot{\omega}_z), \qquad i \in \{x, y\},$$

using the basic arithmetic operators $\{+, -, *\}$. After identifying the best-fitting symbolic expressions, the coefficients were interpreted using a linear regression of the form:

$$a_{\mathrm{lin},i} = c_1\, \omega_z^2 + c_2\, \dot{\omega}_z, \tag{12}$$

from which the sensor coordinates can be extracted as

$$r_x = c_1^{(x)}, \quad r_y = c_1^{(y)}, \quad r_y = c_2^{(x)}, \quad r_x = -c_2^{(y)},$$

corresponding to the structure of Eq. (11).

5.3 Results

The symbolic regression and subsequent coefficient extraction yielded the following coordinates for the acceleration sensor:

$$r_x = (31.84 \pm 0.20) \text{ mm},$$
$$r_y = (10.97 \pm 0.12) \text{ mm}.$$

These results are in excellent agreement with the coordinates determined from the analytical curve fitting method ($r_x = 32.01 \pm 0.17$ mm, $r_y = 11.09 \pm 0.17$ mm), demonstrating that symbolic regression can reliably rediscover the same physical relationships from the measured data without explicit prior modeling. The small discrepancies are within the experimental uncertainties and can be attributed to the stochastic nature of the regression and sensor noise.

5.4 Uncertainty Analysis

Uncertainties were estimated from the covariance matrices of the linear regression performed on the symbolic model coefficients. The diagonal elements of the covariance matrices yield the standard deviations of r_x and r_y:

$$\sigma_{r_i} = \sqrt{C_{ii}}.$$

The resulting relative uncertainties are below 1%, confirming a high level of precision.
The main sources of uncertainty include:

- **Sensor noise:** Random fluctuations in the accelerometer and gyroscope data propagate through both the differentiation of ω_z and the regression model.
- **Model simplification:** The assumption that rotation occurs strictly around the z-axis neglects small tilts or deviations, which may contribute systematic errors on the order of a few tenths of a millimeter.

Overall, the agreement between the symbolic regression and the curve fitting method confirms the robustness of the experimental data and validates the use of data-driven modeling techniques in classical mechanics experiments.

5.5 Measured and Regressed Data

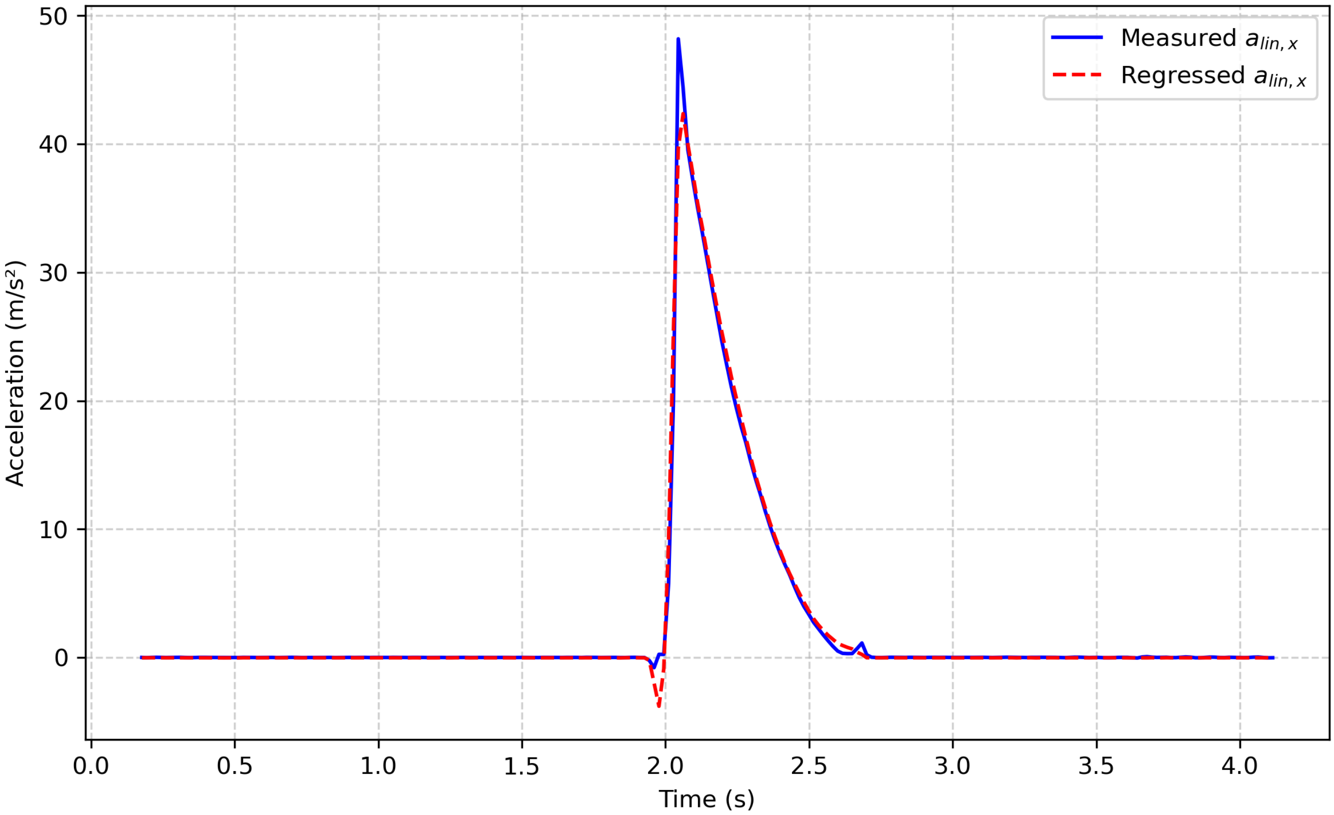

Figure 8: Measured vs Regressed Linear Acceleration (x-component)

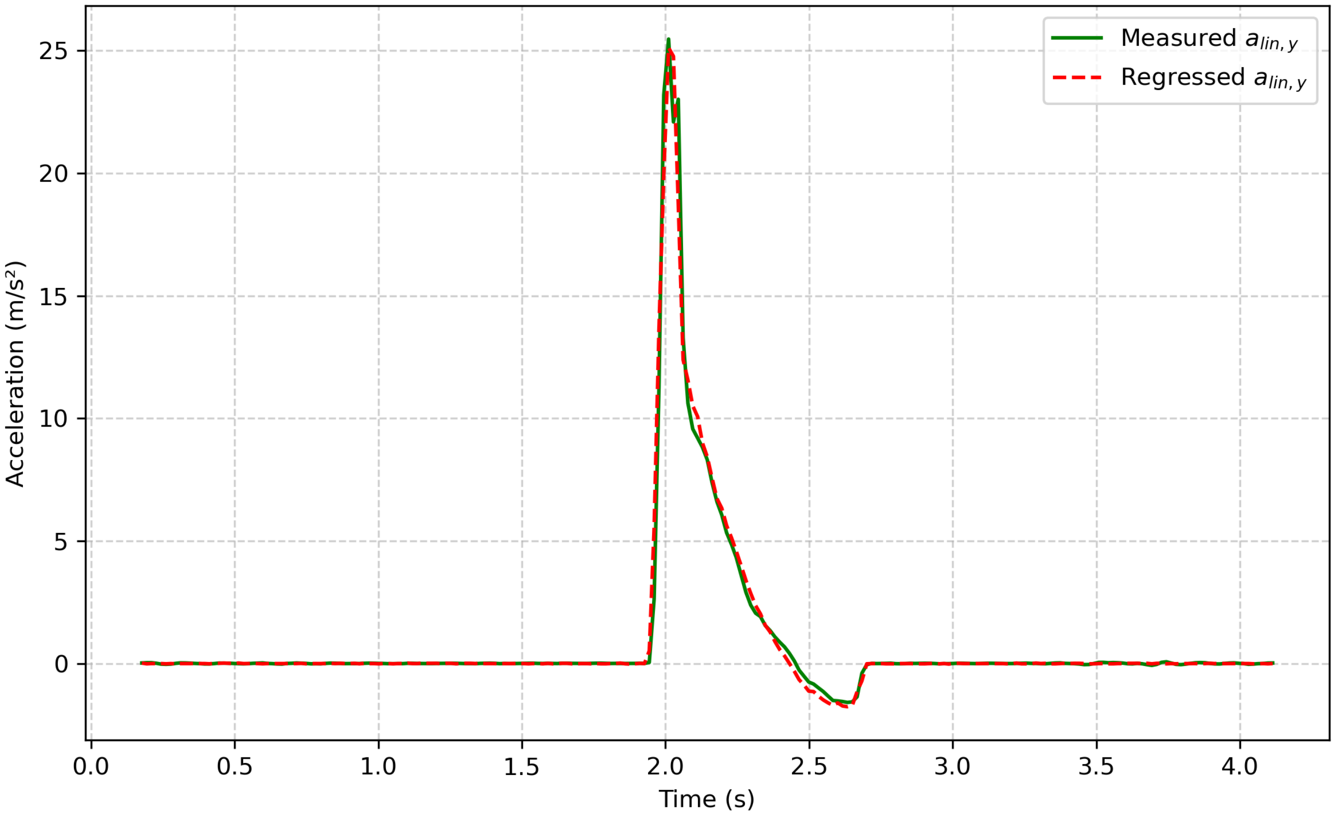

Figure 9: Measured vs Regressed Linear Acceleration (y-component)

Both components show strong agreement between the measured and regressed accelerations. The symbolic regression successfully reconstructed the underlying physical relationships, further confirming the validity of the simplified rotational model and the determined sensor position.

6 Conclusion

In this experiment, the motion of a rotating smartphone was analyzed as an example of an accelerated frame of reference. Using the built-in sensors of a Google Pixel 7 smartphone, the components of angular velocity, acceleration, and linear acceleration were recorded during steady rotation about the vertical z-axis. From these measurements, the position of the internal acceleration sensor relative to the rotation axis was determined by two independent methods: analytical curve fitting and symbolic regression.

The theoretical model, derived from the dynamics of non-inertial frames, relates the measured linear acceleration to the angular velocity and angular acceleration of the rotating smartphone. In the curve fitting approach, the parameters of this model were adjusted using least-squares minimization, yielding the sensor coordinates

$$r_x = (32.01 \pm 0.17)\ \text{mm}, \qquad r_y = (11.09 \pm 0.17)\ \text{mm}.$$

The fitted curves reproduced the measured accelerations with high accuracy, validating the theoretical description. In the symbolic regression approach, a data-driven model discovery algorithm (PySR) was employed to infer the functional relationship between the measured quantities directly from the data. This method rediscovered the same analytical form as the theoretical model and yielded nearly identical results:

$$r_x = (31.84 \pm 0.20)\ \text{mm}, \qquad r_y = (10.97 \pm 0.12)\ \text{mm}.$$

The excellent agreement between the two approaches demonstrates the consistency and reliability of the experimental data and the robustness of both fitting methods. The small discrepancies, well within the combined uncertainties, can be attributed to minor variations in data preprocessing and the stochastic nature of symbolic regression.

The overall uncertainty of the determined sensor position is less than 1%, indicating precise measurements and stable rotational motion. The main sources of uncertainty were sensor noise, numerical differentiation errors, and small deviations from perfect planar rotation. Despite these limitations, both analytical and data-driven methods produced reproducible and physically meaningful results.

In conclusion, the experiment successfully combined theoretical modeling, computational analysis, and modern sensor technology to explore motion in a non-inertial frame. It demonstrated that even everyday devices like smartphones can serve as accurate instruments for physics experiments. The strong correlation between the curve-fitting and symbolic regression results confirms the validity of the theoretical model and highlights the potential of symbolic data analysis in experimental physics.